

2020 12

(PAST PAPER SOLVED)

Q#1

$$x < -1$$

$$x > 1$$

$$(-\infty, -1) \cup (1, \infty)$$

Q#2

$$(i) |x| + |x-1| > 1$$

$$\pm x \pm (x-1) > 1$$

$$x + (x-1) > 1$$

$$-x - (x-1) > 1$$

$$+x - (x-1) > 1, \quad -x + (x-1) > 1$$

are of no use

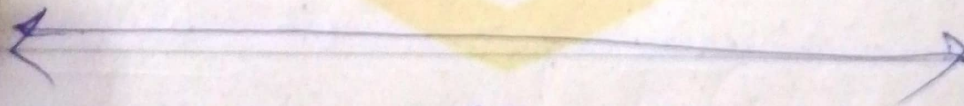
$$2x > 2$$

$$-2x > 0$$

$$x > 1$$

$$x > 0$$

$$\boxed{x > 0} \quad (0, \infty)$$



$$y = e^{ax} \cos(b \tan^{-1} x)$$

$$\frac{dy}{dx} = e^{ax} \frac{d}{dx} \cos(b \tan^{-1} x) + \cos(b \tan^{-1} x) \frac{d}{dx} e^{ax}$$

$$= e^{ax} (-\sin b \tan^{-1} x) \frac{d}{dx} b \tan^{-1} x + \cos(b \tan^{-1} x) a e^{ax}$$

$$= e^{ax} \left[a \cos b \tan^{-1} x - \sin b \tan^{-1} x \times b \right]$$

$$= e^{ax} \left[a \cos b \tan^{-1} x - b \sin b \tan^{-1} x \right]$$

$$y = \left[x \cdot a^x \cdot \sinh x \right]$$

$$\frac{dy}{dx} = \frac{d}{dx} \left[x \cdot a^x \cdot \sinh x \right]$$

$$= x \cdot a^x \frac{d}{dx} \sinh x + x \cdot \sinh x \frac{d}{dx} a^x + a^x \sinh x \frac{d}{dx} x$$

$$= x a^x \cosh x + x \sinh x [a^x \ln a] + a^x \sinh x \cdot 1$$

$$\frac{dy}{dx} = a^x [x \cosh x + x \ln a \sinh x + \sinh x]$$

$$e^{ax} \frac{d}{dx} (ax) = a e^{ax} \frac{d}{dx} x$$

$$y = \sin(\sqrt{1+\cos x})$$

$$\frac{dy}{dx} = \frac{d}{dx} \sin(\sqrt{1+\cos x})$$

$$= \cos(\sqrt{1+\cos x}) \frac{d}{dx}(\sqrt{1+\cos x})$$

$$= \cos(\sqrt{1+\cos x}) \frac{1}{2\sqrt{1+\cos x}} \frac{d}{dx}(1+\cos x)$$

$$= \cos(\sqrt{1+\cos x}) \frac{(-\sin x)}{2\sqrt{1+\cos x}}$$

$$= \frac{(-\sin x)(\cos \sqrt{1+\cos x})}{2\sqrt{1+\cos x}}$$

$$(b) \ y = x \ln x \cos^2 \pi x$$

$$\frac{dy}{dx} = x \ln x \frac{d}{dx} \cos^2 \pi x$$

$$+ x \cos^2 \pi x \frac{d}{dx} \ln x$$

$$+ \ln x \cos^2 \pi x \frac{d}{dx} x$$

$$= x \ln 2 \cos^2 x \frac{d}{dx} \cos^2 x$$

$$+ x \cos^2 x \times \frac{1}{x} + (\ln \cos^2 x) (1)$$

$$= x \ln 2 \cos^2 x (-\sin^2 x) \frac{1}{x} + \cos^2 x + \ln \cos^2 x$$

$$\frac{dy}{dx} = \cos^2 x (1 + \ln 2) - 2x \ln \cos^2 x \sin^2 x$$

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$

$$\text{let } \sqrt{x} = t$$

$$\frac{1}{2\sqrt{x}} dx = dt$$

$$\frac{1}{\sqrt{x}} dx = 2dt$$

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \int (\sin t) (2dt)$$

$$= -2 \cos t + C$$

$$= C - 2 \cos \sqrt{x} \quad \text{Ans}$$

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = C - 2 \cos \sqrt{x}$$

$$\int \frac{1}{3\sin x + 4\cos x} dx$$

$$\text{let } r \sin(x+\phi) = 3\sin x + 4\cos x$$

$$r \sin x \cos \phi + r \cos x \sin \phi = 3\sin x + 4\cos x$$

$$r \cos \phi = 3 \quad r \sin \phi = 4$$

$$r^2 \cos^2 \phi + r^2 \sin^2 \phi = 3^2 + 4^2$$

$$r^2 (\cos^2 \phi + \sin^2 \phi) = 25$$

$$r^2 = 25$$

$$r = 5$$

$$3\sin x + 4\cos x = 5 \sin(x+\phi)$$

$$\int \frac{1}{3\sin x + 4\cos x} = \int \frac{1}{5\sin(x+\phi)} dx$$

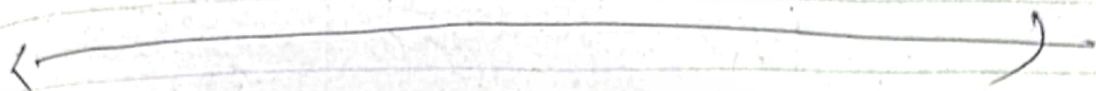
$$= \frac{1}{5} \int \operatorname{cosec}(x+\phi) dx$$

$$= +\frac{1}{5} \ln | \csc(\alpha+\phi) - \cot(\alpha+\phi) |$$

$$\frac{\sin \phi}{\cos \phi} = \frac{4}{3}$$

$$\tan \phi = \frac{4}{3}$$

$$\phi = \tan^{-1}\left(\frac{4}{3}\right)$$



$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$\text{Let } a^x - 1 = y$$

as $x \rightarrow 0$

$$a^x = (1+y)$$

$$\text{as } x \rightarrow 0 \quad y \rightarrow a^0 - 1 = 1 - 1 = 0$$

Sol

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$\ln a^x = \ln(1+y)$$

$$x \ln a = \ln(1+y)$$

$$x = \frac{\ln(1+y)}{\ln a}$$

$$\text{Let } a^x - 1 = y$$

$$\text{as } x \rightarrow 0 \quad y = a^0 - 1 = 1 - 1 = 0$$

$$a^x = (1+y)$$

$$\text{So } \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{y \rightarrow 0} \frac{y}{\frac{\ln(1+y)}{\ln a}}$$

$$= \lim_{y \rightarrow 0} \frac{y}{\ln(1+y)} \cdot \ln a$$

So

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{y \rightarrow 0} \frac{y}{\ln(1+y)}$$

$$= \lim_{y \rightarrow 0} \ln a \left[\frac{1}{\frac{1}{y} \ln(1+y)} \right]$$

$$= \lim_{y \rightarrow 0} \ln a \left[\frac{1}{\frac{1}{y} \ln(1+y)} \right]$$

$$= \ln a$$

$$\lim_{y \rightarrow 0} \left[\frac{1}{\ln(1+y) \frac{1}{y}} \right]$$

$$= \ln a$$

$$\ln \left[\lim_{y \rightarrow 0} (1+y)^{\frac{1}{y}} \right]$$

$$= \frac{\ln a}{\ln e}$$

$$= \frac{\ln a}{1} = \ln a$$

$$y = x^x e^x \sin x (\ln x)$$

$$\frac{d}{dx} x^x = x^x (1 + \ln x)$$

$$\ln y = x \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = x \cdot \frac{1}{x} + \ln x (1)$$

$$\frac{dy}{dx} = x^x (1 + \ln x)$$

Let $y = x^x$

$$\ln y = \ln x^x = x \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = x \frac{d}{dx} \ln x + \ln x \frac{dx}{dx}$$

$$= x \cdot \frac{1}{x} + (\ln x) (1)$$

$$\frac{dy}{dx} = y (1 + \ln x)$$

$$= x^x (1 + \ln x)$$

$$\text{So } \boxed{\frac{d}{dx} x^x = x^x (1 + \ln x)}$$

Now

$$y = x^x e^x \sin x (\ln x)$$

$$\frac{dy}{dx} = \frac{d}{dx} [x^x e^x \sin x \ln x]$$

$$= x^x e^x \sin x \frac{d}{dx} \ln x$$

$$+ x^x e^x \ln x \frac{d}{dx} \sin x$$

$$+ x^x \ln x \sin x \frac{d}{dx} e^x$$

$$+ e^x \ln x \sin x \frac{d}{dx} x^x$$

$$= x^x e^x \sin x \times \frac{1}{x}$$

$$+ x^x e^x \ln x \cos x$$

$$+ x^x \ln x \sin x e^x$$

$$+ e^x \ln x \sin x [x^x (1 + \ln x)]$$